

Nishit Kumar

Associate Technical Consultant

SOFTWARE · DATA · AI

Building practical software systems by combining software engineering, data, machine learning, and AI from OCR-driven product analysis to adaptive speech research.

graduate



software · data · ai



engineering systems



projects



solutions

summary

Nishit Kumar is a Computer Science & Engineering graduate (Amrita Vishwa Vidyapeetham, Bengaluru) currently working as an **Associate Technical Consultant** at Perficient. His work spans the full software lifecycle — planning, architecture, backend logic, and frontend delivery — with applied experience across data science, machine learning, generative AI, and OCR-based intelligent applications. He has shipped full-stack platforms on Angular, Node.js, and MongoDB, an AI product advisor that reads packaged-food ingredient lists from a photo, and a generative AI prototype with speech input. His final-year research focuses on an adaptive, emotion-aware speech therapy framework for dysarthria. He writes documented, testable code and treats every system as an engineering problem to be measured — not merely solved.

technical_strengths

- 01

End-to-end delivery
— from data pipelines and APIs to polished frontends.
- 02

Applied AI & OCR — turning unstructured input (labels, speech, logs) into decisions.
- 03

Research thinking — framing problems, designing methods, measuring outcomes.
- 04

Clear communication — technical explanations that non-specialists can act on.

career_direction

Deepening expertise in products where software, data, and AI meet intelligent applications, applied machine learning, and systems engineering while growing into technical roles that combine deep problem-solving with client and team-facing communication.

skills_overview

programming

Python

Java

C

JavaScript

frontend

HTML

CSS

Angular

Tailwind CSS

backend

Node.js

Express.js

Flask

databases

MySQL

MongoDB

data & ml

Pandas

NumPy

Scikit-learn

XGBoost

LightGBM

ai & ocr

Machine Learning

Generative AI

Mistral AI

Tesseract OCR

tools

Git

GitHub

VS Code

Postman

2026 – present

Associate Technical Consultant

Perficient

Current role — technical consulting across client engagements. *[responsibilities can be added here as they evolve]*

2025

Full Stack Developer Intern

Kekkar Infotech & Services

Developed a hospital appointment & surgery information platform with a modern web stack:

Angular

Node.js

MongoDB

Maps integration

Scope included clinic search, appointment booking, surgery information pages, downloadable PDF generation, and embedded maps.

engagement_focus

Across internships and client work, the recurring pattern is the same: understand the domain, map the data, design the flow, then build and explain it — **business + engineering together**.

problem

Ingredient lists are dense and unfamiliar, and generic food advice ignores the reader's own allergens, conditions, or dietary choices — so a quick in-store decision stays hard.

solution

Photograph an ingredient list, extract the text with OCR, evaluate every ingredient against a personal dietary profile with a large language model, and return an explainable verdict with safer alternatives.

architecture processing_pipeline



technology

PHP · Tailwind CSS · Python · Flask · Tesseract OCR · Mistral AI · MySQL

key features

User registration · personalized dietary profile · ingredient scanning · OCR processing · AI-powered analysis · product classification · explainable verdicts · alternative recommendations · decision history.

contribution

Implemented the end-to-end pipeline — image handling into Tesseract OCR, ingredient structuring, the Mistral AI analysis layer, and personalized Safe / Caution / Avoid classification with explanation generation. Built the alternative-recommendation logic and wired decision history into the database layer.

outcome

A working end-to-end prototype that turns an unreadable photo of an ingredient list into a clear, explainable, personal recommendation — a production-shaped OCR-to-LLM flow rather than a loose chain of scripts.

Adaptive Emotion-Aware Speech Therapy Framework for Dysarthria Using AI

research_problem

Conventional dysarthria therapy applies fixed exercise routines, ignoring each speaker's unique impairment profile — severity, accent variation, and emotional state during a session.

methodology

Input speech is preprocessed and reduced to acoustic features. Models classify speech characteristics, emotion, and severity; the assessment feeds an adaptive therapy generator that stages exercises; feedback updates the next round.

speech_therapy_loop



contribution

Defined the adaptive therapy decision layer that connects speech and emotion analysis to severity-appropriate exercise generation — turning a one-size-fits-all therapy model into a measurable, feedback-driven system.

dataset context

Developed as a research framework; evaluation targets standard dysarthric and emotion speech corpora in the domain.

motivation

Therapy is more effective when it adapts. Continuously reading speech and emotion signals lets exercises stay challenging but achievable, and adjust when the speaker is struggling.

ai / ml approach

Machine and deep learning classifiers over extracted speech features for emotion and severity estimation, accent-aware feature handling, and a therapy decision layer that turns model output into staged exercise plans.

`full_stack_development internship`

Hospital Appointment & Surgery Information Platform

Patient-facing platform for clinic discovery, appointment booking, surgery information, downloadable PDFs, and embedded maps. Developed during a Full Stack Developer internship.

Clinic Search → Booking → Surgery Info → PDF + Maps

`angular · node.js · mongodb``generative_ai hackathon`

Text-to-Image Generation System

Hackathon-built generative AI prototype that turns text or spoken input into a generated image through a clean chatbot-style frontend, making AI interaction approachable and immediate.

Text / Speech → Processing → AI Model → Generated Image

`generative ai · speech recognition · ui/ux``systems · machine_learning`

Log-Based Predictive Autoscaler

Predictive autoscaling system that consumes system and application logs, extracts behavioral features, and forecasts resource demand — adjusting infrastructure before saturation instead of after.

Logs → Feature Extraction → Prediction Model → Resource Demand → Autoscaling Decision

`python · machine learning · systems`

Technical Profile

programming	Python Java C JavaScript
frontend	HTML CSS Angular Tailwind CSS
backend	Node.js Express.js Flask
data & ml	Pandas NumPy Scikit-learn XGBoost LightGBM
ai & ocr	Data Science Machine Learning Generative AI Mistral AI Tesseract OCR
databases	MySQL MongoDB
tools	Git GitHub VS Code Postman

education

2022 – 2026

B.Tech — Computer Science & Engineering

Amrita Vishwa Vidyapeetham, Bengaluru

certifications

Python Programming & Data Analytics

Udemy

hackathons

NNDL Hackathon

Text-to-image generation system — frontend/UI, chatbot-style AI interaction, speech recognition integration.

`/contact`

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